Learned Complementary Preconditioning for Sparse RBF–FD Discretizations

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Abstract

This report studies the inverse action left unresolved by strong classical preconditioning, $d(r) = A^{-1}r - M^{-1}r$, for sparse polynomially augmented RBF–FD discretizations. Strong ILU already solves the tested Poisson, scalar-contrast, moderate node-irregularity, and scale-only systems efficiently. Rotated anisotropic diffusion creates an accurate but difficult regime in which system-own Ritz spaces can accelerate nonsymmetric recycled GMRES. Direct learned corrections did not outperform ILU. Learned subspace prediction became comparable with deterministic interpolation only within a narrow spatially constant coefficient family.

For spatially varying tensor fields, short Arnoldi constructions initially failed. Deeper trajectories and adequate retained dimension recovered an own-system union oracle on all 15 original hard fields, and rank adaptation was validated on 17 independently sampled hard systems. These successes did not yield reliable cross-field reuse. Training-dictionary truncation discarded solver-relevant information, while neural selection remained poor even when training targets were representable. Target-local probe construction recovered most teacher benefit on six of seven later hard targets once rank 48 was retained, but left a collective missing subspace on the remaining target.

Raw multi-RHS Krylov information contained a useful rank-12 correction for that failure. Its teacher-informed extraction reduced the difficult solve from the 300-step cap to 92 iterations. Neural mixing and a subsequently tested equivariant operator family failed to recover this effect. Direct projector analysis confirmed a large representation deficit in the tested operator span. The primary open problem is neural selection from a reusable dictionary using both the coefficient field and a very short target-local probe. The network must improve the probe-cost and solver-performance tradeoff relative to classical selection at the same probe depth.

1 Problem and numerical framework

The study asks whether a strong RBF–FD-specific preconditioner leaves a reproducible inverse action that learning can exploit. A direct model may learn the full missing action without assuming low dimension. A subspace model becomes meaningful only after a compact, stable, solver-relevant difficult space has been demonstrated. Proper orthogonal decomposition (POD) therefore serves as a diagnostic and classical reference. This framework defines the alternatives tested throughout the report.

The complete computational framework is summarized here before the empirical development. Consider

$$-\nabla\cdot\big(K(x)\nabla u(x)\big) = f(x), \qquad x \in \Omega = [0,1]^2, \qquad u = g \text{ on } \partial\Omega, \tag{1}$$

where $K = I$ gives Poisson, $K = a(x)I$ gives scalar diffusion, and K may be a symmetric positive-definite tensor. Manufactured sine-series solutions provide analytic forcing and boundary data. Thus discretization error and linear-solver residual are measured separately.

1.1 Local RBF–FD operator

The established experiments use quasi-uniform tensor-product nodes with fixed boundary nodes and reproducible interior jitter. At each interior point x_i, a k-d tree returns the s nearest nodes S_i. The stencil is translated by x_i and divided by its farthest-node radius ρ_i. With the polyharmonic spline

$$\phi(r) = r^3 \tag{2}$$

and all monomials of total degree at most two, $p = (1, x, y, x^2, xy, y^2)$, the local weights solve

$$\begin{bmatrix} \Phi_i & P_i \\ P_i^{\mathsf{T}} & 0 \end{bmatrix} \begin{bmatrix} w_i \\ \lambda_i \end{bmatrix} = \begin{bmatrix} \mathcal{L}_i \phi(\|\cdot - x_j\|)|_{x_i} \\ \mathcal{L}_i p|_{x_i} \end{bmatrix}. \tag{3}$$

Here $(\Phi_i)_{jk} = \phi(\|\widehat{x}_j - \widehat{x}_k\|)$ and P_i evaluates the scaled monomials. First derivatives are rescaled by ρ_i^{-1} and second derivatives by ρ_i^{-2}. No regularization or silent least-squares fallback is used. A failed local augmented solve is recorded as an invalid node set.

For constant K, $\mathcal{L}u = -K_{11}u_{xx} - 2K_{12}u_{xy} - K_{22}u_{yy}$. For a spatial tensor, the implementation includes the product-rule drift exactly,

$$\mathcal{L}u = -K : \nabla^2 u - (\nabla \cdot K) \cdot \nabla u. \tag{4}$$

Each interior row inserts w_i only at stencil columns. A boundary row is the corresponding identity row and its right-hand side is the manufactured Dirichlet value. The result is the sparse, generally nonsymmetric system $Au = b$.

1.2 Classical and complementary inverse actions

The principal baseline is SuperLU threshold ILU with drop tolerance 10^{-3}, fill factor 5, COLAMD ordering, and zero diagonal-pivot threshold. It defines a fixed linear inverse action M^{-1} [7]. An early RBF–FD-specific control assembles a cheaper matrix B_s on the same nodes with a smaller stencil and applies $\mathrm{ILU}(B_s)^{-1}$ [4]. Later experiments retain the stronger $\mathrm{ILU}(A)$ baseline.

For any residual or Arnoldi vector r, the exact missing inverse action is

$$d(r) = A^{-1}r - M^{-1}r. \tag{5}$$

The term $A^{-1}r$ is always obtained from an accurate solve, never by forming an inverse. With training snapshots $D = [d^{(1)}, \ldots, d^{(K)}] = U\Sigma V^{\mathsf{T}}$, the POD basis is $Z_q = U_{:,1:q}$. Its captured energy and held-out error are

$$E_q = \frac{\sum_{i=1}^{q} \sigma_i^2}{\sum_i \sigma_i^2}, \qquad \eta_q = \frac{\left\|(I - Z_q Z_q^{\mathsf{T}}) d_{\mathrm{test}}\right\|_2}{\|d_{\mathrm{test}}\|_2}. \tag{6}$$

Captured energy alone does not establish solver acceleration. Every proposed space is tested inside the Krylov method.

All hard-regime solvers use right preconditioning. Arnoldi is applied to

$$B = AM^{-1}, \qquad BV_m = V_{m+1}\overline{H}_m. \tag{7}$$

Ordinary Ritz pairs are eigenpairs (λ_i, g_i) of the square part H_m. The residual-coordinate Ritz direction is $y_i = V_m g_i$, and its solution-coordinate direction is

$$z_i = M^{-1} y_i. \tag{8}$$

The default oracle selects the smallest $|\lambda_i|$, expands a nonreal vector into its real and imaginary directions, maps each direction with M^{-1}, and twice reorthogonalizes in selection order. Thus

$$Z = \mathrm{orth}\big([M^{-1}y_1, \ldots, M^{-1}y_q]\big). \tag{9}$$

Unless stated otherwise, the fixed anisotropy studies use 64 Arnoldi steps and $q = 8$.
Because A is nonsymmetric, the correct paired construction is

$$AZ = WR, \qquad W^{\mathsf{T}}W = I, \qquad U = ZR^{-1}, \qquad AU = W, \tag{10}$$

where the thin QR factorization produces W, R and U is formed by a triangular solve. The multiplicative Petrov–Galerkin preconditioner first computes $y_0 = M^{-1}r$ and $r_1 = r - Ay_0$, then applies

$$P_{\mathrm{PG}}(r) = M^{-1}r + ZR^{-1}W^{\mathsf{T}}r_1. \tag{11}$$

The algebraically equivalent correction is $UW^{\mathsf{T}}r_1$.
The final oracle integration uses fixed-space recycled GMRES in the GCRO family [2, 6]. At each restart it forms

$$\bar{x} = x + UW^{\mathsf{T}}r, \qquad \bar{r} = (I - WW^{\mathsf{T}})r, \tag{12}$$

then applies Arnoldi to $(I - WW^{\mathsf{T}})AM^{-1}$. If $Y_m = M^{-1}V_m$ and $G_m = W^{\mathsf{T}}AY_m$, the trial correction is $(Y_m - UG_m)y$. This compensation is essential because $A(Y_m - UG_m) = (I - WW^{\mathsf{T}})AY_m$. The pair (U, W) is held fixed across restarts, which isolates the supplied Z from adaptive updates.
The learned recycle-space experiments represent Z in a training-only dictionary $\Psi \in \mathbb{R}^{N\times p}$ obtained from an SVD of oracle snapshots. A small MLP predicts a coefficient matrix $C_\theta(\xi) \in \mathbb{R}^{p\times q}$, after which

$$\xi \longmapsto C_\theta(\xi) \longmapsto Z_{\mathrm{raw}} = \Psi C_\theta(\xi) \longmapsto Z_\theta = \mathrm{orth}(Z_{\mathrm{raw}}). \tag{13}$$

The target-side test space is $W_\theta = \mathrm{orth}(A(\xi)Z_\theta)$. The principal supervised objective is the basis-invariant residual-space loss

$$L_W = \left\| W_\theta W_\theta^{\mathsf{T}} - W_* W_*^{\mathsf{T}} \right\|_F^2. \tag{14}$$

Nearest-oracle reuse and Grassmann geodesic interpolation are deterministic controls [1, 3]. A learned method is credited only when it improves the held-out recycled solve after setup and prediction costs are exposed.

2 Discretization validation and hard-regime discovery

The preceding section defines the operator and complementary actions. Before testing learning, the discretization and the strength of the classical baseline must be established. The first verification used $\phi(r) = r^3$, polynomial degree two, $s = 25$, and uniform grids from $N = 100$ to 484. The exact solution was $u = \sin(\pi x)\sin(\pi y)$. The direct residual remained below 10^{-13} and the final refinement step had observed error rate 1.49. The matrix became sparser with refinement while remaining visibly nonsymmetric. Table 1 and figure 1 give the independent discretization evidence.

Table 1: Poisson RBF–FD refinement. Condition numbers are feasible dense 2-norm estimates for these small verification systems.

N	nnz(A)	$\|A - A^T\|_F / \|A\|_F$	$\kappa_2(A)$	relative error
100	1,636	0.253	3.81×10^3	2.78×10^{-2}
196	3,652	0.203	9.25×10^3	1.89×10^{-2}
324	6,468	0.174	1.77×10^4	1.28×10^{-2}
484	10,084	0.155	2.96×10^4	9.31×10^{-3}

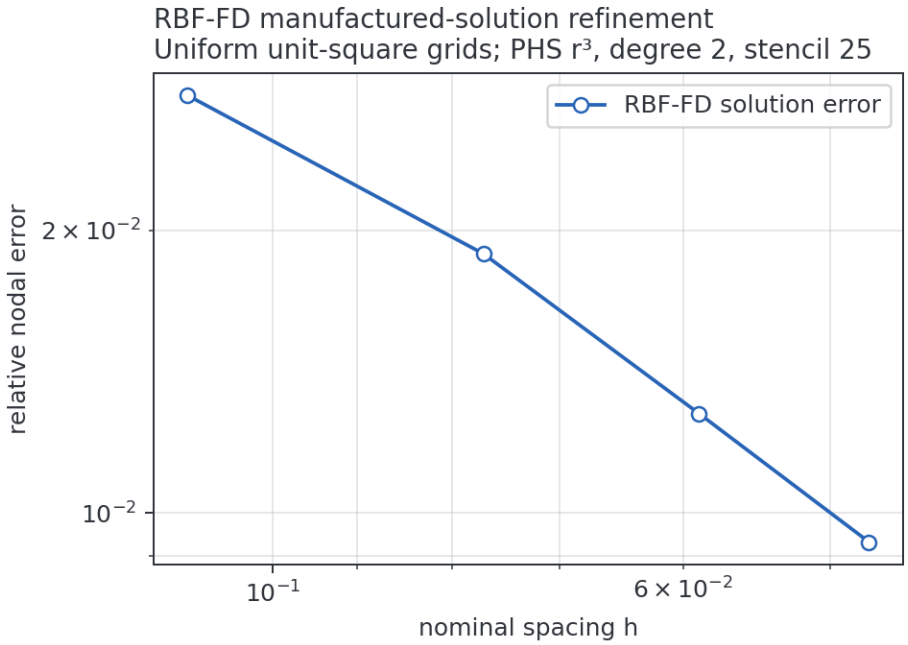


Figure 1: Manufactured-solution convergence for the verified Poisson discretization.

The first preconditioning system used $N = 324$, the same $s = 25$ stencil, seed-7 jitter of 0.15, restart 40, tolerance 10^{-9}, and a 500-step cap. Its relative discretization error was 5.57×10^{-3}, its matrix had 6,468 nonzeros, and its symmetry defect was 0.180. Table 2 shows why ILU had to remain the reference. The smaller-stencil construction reduced iterations but its prototype assembly and factorization cost dominated the solve.

Table 2: Classical baselines on the $N = 324$ jittered system.

Method	Iterations	Setup (s)	Solve (s)	Total (s)
No preconditioner	122	0	0.0327	0.0327
ILU(A)	5	0.00172	0.000763	0.00248
Smaller-stencil B_{13} with ILU(B_{13})	12	0.0385	0.00128	0.0397

The next hierarchy varied resolution, jitter, stencil size, scalar coefficient contrast, and ILU budget one factor at a time. Across 15 systems with $N = 93$–800, strong ILU required only 5–11 iterations. This was an important negative result because a learned complement had no realistic room to improve most early tests. Scaling to $N = 20{,}164$ preserved that conclusion. At the largest size, A occupied 5.691 MiB, ILU occupied 15.289 MiB with fill ratio 2.696, setup took 0.299 s, and the median solve required 19.5 iterations. Assembly, at 3.294 s, was the dominant one-system cost.

Table 3: Scale-only RBF–FD systems with $s = 25$, seed-7 jitter 0.15, and unchanged strong ILU.

N	nnz	A MiB	assembly (s)	ILU setup (s)	ILU MiB	fill	iterations
2,500	57,796	0.671	0.386	0.024	1.594	2.380	9.0
5,041	119,305	1.385	0.798	0.050	3.588	2.600	11.0
10,000	240,496	2.790	1.594	0.121	7.481	2.690	14.0
20,164	490,564	5.691	3.294	0.299	15.289	2.696	19.5

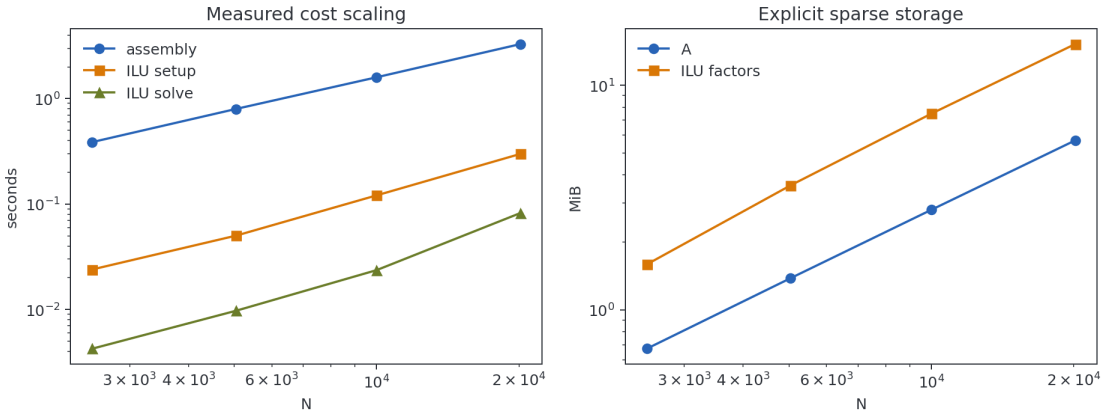


Figure 2: Assembly, ILU setup, online solve, and sparse-storage scaling. No scale-only ILU memory wall appears by $N \simeq 20{,}000$.

Smooth scalar contrast through 10^4, a localized contrast-10^3 inclusion, and valid centre refinement also remained easy. In contrast, the standard SPD tensor obtained by rotating eigenvalues $(1, R)$ by 30° created a sharp transition. At $R = 100$ and 1000, ILU reached the 300-step cap even though the direct manufactured-solution errors were 0.0142 and 0.00153. This was the first regime in which the discretization remained meaningful while the strong classical solve was genuinely difficult.

Table 4: PDE difficulty at $N = 5{,}041$. A reported value of 300 is a cap hit, not an inferred converged count.

System	Contrast	Symmetry defect	Conditioning proxy	ILU iterations	Solve (s)	Disc. error
Smooth scalar	10	0.087	2.64×10^2	11.0	0.009	5.48×10^{-4}
Smooth scalar	10^4	0.078	1.81×10^4	10.5	0.008	2.0×10^{-3}
Local scalar	10^3	0.069	5.05×10^6	11.0	0.011	1.2×10^{-2}
Rotated tensor	$R = 10$	0.129	1.23×10^4	10.0	0.008	5.57×10^{-4}
Rotated tensor	$R = 100$	0.139	4.51×10^7	300 cap	0.397	1.42×10^{-2}
Rotated tensor	$R = 1000$	0.140	2.08×10^8	300 cap	0.459	1.53×10^{-3}

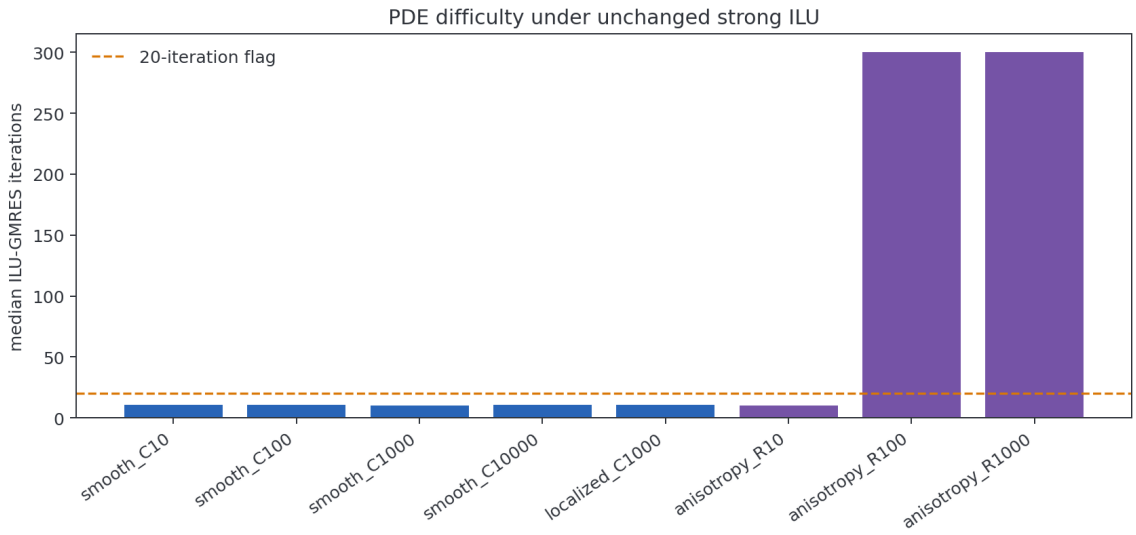


Figure 3: Strong-ILU behavior under scalar contrast, localized contrast, and rotated anisotropy.

3 Residual action and first recycle-space evidence

The validated discretization revealed that strong ILU already solved most regimes cheaply. The next experiments ask what inverse action remains unresolved and whether it has a useful low-dimensional structure. On the initial easy Poisson system, 48 training corrections and 24 held-out corrections mixed synthetic samples with actual GMRES histories. At $q = 5$, POD captured 96.5% of snapshot energy, while mean held-out error was 0.211. At $q = 8$, energy reached 99.0% and mean error fell to 0.179. The classical solve already required only five iterations. POD at $q = 20$ reduced one representative solve to four, while a random rank-20 control remained at five. The added setup could not improve total time. Approximate low rank alone did not create a useful preconditioner.

The broader one-factor study reinforced the distinction. Joint correction snapshots often had $q_{95} \leq 4$, yet held-out errors reached 0.584 and fifteen high-energy cases produced no Krylov gain. Across those systems, captured energy and actual iteration reduction had Spearman correlation 0.745, but high energy alone was not sufficient. Figure 4 shows the normalized spectra, captured energy, and held-out projection curves on the harder family.

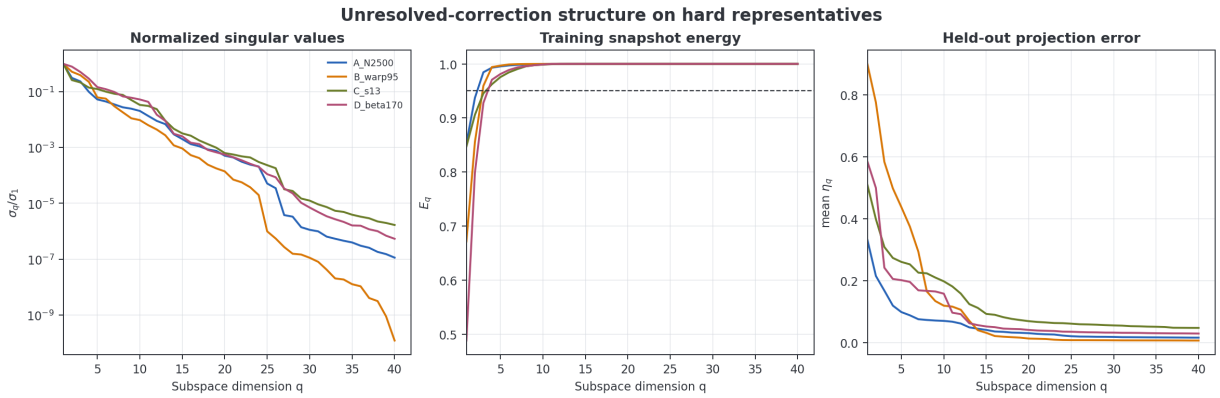


Figure 4: Unresolved-correction POD diagnostics. Held-out projection, rather than training energy alone, determines whether low-dimensional evidence is credible.

When the systems became large or anisotropic, low dimensionality ceased to persist uniformly. Several training snapshot matrices reached 95% energy with two to five modes, while only the $R = 100$ anisotropy case passed the held-out criterion. The $R = 1000$ case had $q_{95} = 3$ with held-out error 0.790. Scale $N = 20{,}164$ had $q_{95} = 5$ with held-out error 0.439. The tiny ratios q/N therefore failed to identify an exploitable space.

Table 5: Correction-POD dimensionality on selected nontrivial or setup-sensitive systems.

System	N	q_{95}	q_{95}/N	held-out η_{95}	Structured at 95%?
Scale $N = 20{,}164$	20,164	5	2.48×10^{-4}	0.439	no
Smooth scalar 10^4	5,041	2	3.97×10^{-4}	0.478	no
Centre refined	5,041	2	3.97×10^{-4}	0.837	no
Rotated anisotropy $R = 100$	5,041	2	3.97×10^{-4}	0.041	yes
Rotated anisotropy $R = 1000$	5,041	3	5.95×10^{-4}	0.790	no

POD was next compared with Krylov-derived spaces. On the scale-only $N = 20{,}164$ system, a rank-16 ordinary Ritz oracle reduced 20 iterations to 10. Smooth scalar contrast 10^4 improved from 11 to 6 and a valid centre-refined system from 12 to 7. These iteration reductions began from easy solves and did not justify learning. The decisive result was rotated anisotropy at $R = 100$: rank-24 correction POD reduced the capped baseline to 99 iterations, whereas rank-24 Ritz reduced it to 48, an 84% observed reduction. At $R = 1000$, neither construction converged by rank 24. Ritz information became the main diagnostic target because it was more solver relevant in the one hard positive regime.

4 Paired projection for nonsymmetric systems

The previous section found the first decisive recycle-space benefit in rotated anisotropy. Using that space correctly requires a nonsymmetric construction and a fair comparison of persistent correction with Krylov recycling. Across $R = 85$–120, $\left\|A - A^{\mathsf{T}}\right\|_F / \|A\|_F$ was 0.138–0.139. A 48-probe estimate for the right-preconditioned operator gave $\left\|AM^{-1} - (AM^{-1})^{\mathsf{T}}\right\|_F / \left\|AM^{-1}\right\|_F$ between 1.297 and 1.625. A later verification confirmed that the hard-regime code already used the $AZ = WR$ minimum-residual Petrov construction. Rebuilt rank-eight actions reproduced all earlier iterations exactly, and the historical, explicit Petrov, and recycling-normalized actions agreed to relative error at most 2.29×10^{-15}.

The fixed protocol used $N = 5{,}041$, $s = 25$, node seed 7, jitter 0.15, rotation 30°, ILU drop tolerance 10^{-3} and fill 5, restart 50, tolerance 10^{-8}, and cap 300. The oracle came from a 64-step construction trajectory with seed 20261001 and was evaluated on RHS seeds 20261101–20261103. Table 6 and figure 5 compare persistent Petrov correction with fixed-space recycled GMRES using exactly the same Z. Iterations are identical at every R. Recycling avoids a reduced solve at every preconditioner application and has the lower median online time, with persistent/recycling ratio 1.26.

Table 6: Same-system rank-eight oracle comparison. Times are median online wall-clock measurements.

R	ILU	Persistent PG	Recycling	PG time (s)	Recycle time (s)	Convergence
85	87	32	32	0.041	0.037	1.00
90	148	35	35	0.051	0.049	1.00
95	148	36	36	0.051	0.045	1.00
100	300	78	78	0.114	0.087	1.00
105	290	71	71	0.102	0.081	1.00
110	300	84	84	0.147	0.100	1.00
120	300	298	298	0.468	0.326	1.00

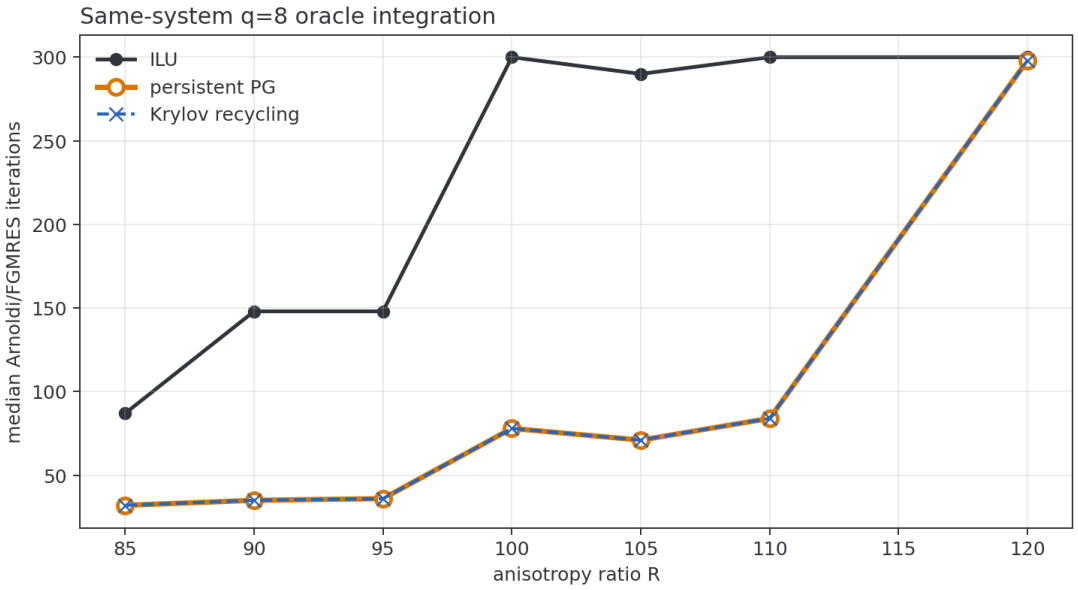


Figure 5: Persistent Petrov–Galerkin and recycled GMRES exploit the same oracle information and match in iteration count.

The residual-space component explains why one-time correction was insufficient. In the persistent method, the median $\|W^{\mathsf{T}} r_1\| / \|r_1\|$ after the first application was 0.454 at $R = 90$, 0.485 at $R = 100$, and 0.669 at $R = 110$. Each coarse correction reduced it to roundoff. Projected Arnoldi kept the corresponding true-residual component near 4.22×10^{-10} in median. Applying the persistent correction only every fifth preconditioner call cost 14 extra median iterations, and applying it only at initialization cost 121. Repeated enforcement is material, and the Krylov projector supplies it more efficiently.

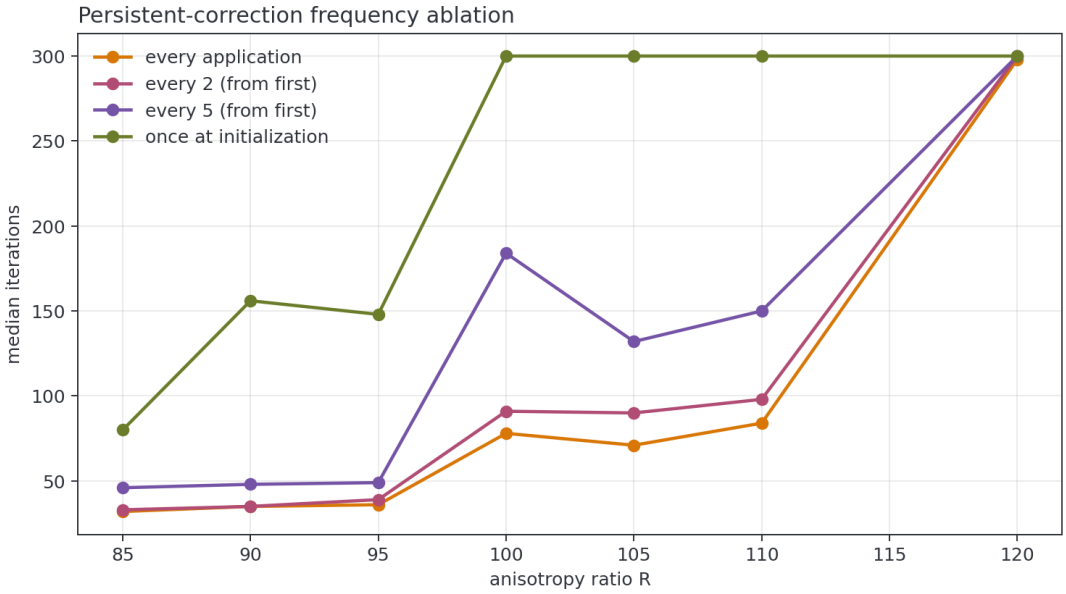


Figure 6: Frequency ablation for the rank-eight Petrov correction. A single initialization does not reproduce persistent enforcement.

This relationship is close in spirit to the projected invariant-space mechanism of NeurKItt [5]: a conventional preconditioner remains active while a predicted space guides Krylov projection. The RBF–FD question asks whether the space represents a persistent low-dimensional failure of a strong, application-specific ILU and can be predicted or updated economically. The scientific contribution must therefore lie in identifying and predicting that failure space.

5 Transferability of the first Ritz oracle

The paired recycle method can exploit an effective oracle space. The unresolved question is whether the first rank-eight Ritz spaces form stable targets across nearby systems. Fine sampling over $R = 60$–140 exposed a transition rather than a uniformly hard family. Own-system medians included $148 \to 35$ at $R = 90$, $300 \to 78$ at $R = 100$, and $300 \to 84$ at $R = 110$. At $R = 120$, the same construction gave only $300 \to 298$, and it failed at 130 and 140. Across the hard ratios, median reduction was 65.9% and convergence was 0.83. Figure 7 shows that the solver-critical dimension and onset of benefit depend on R.

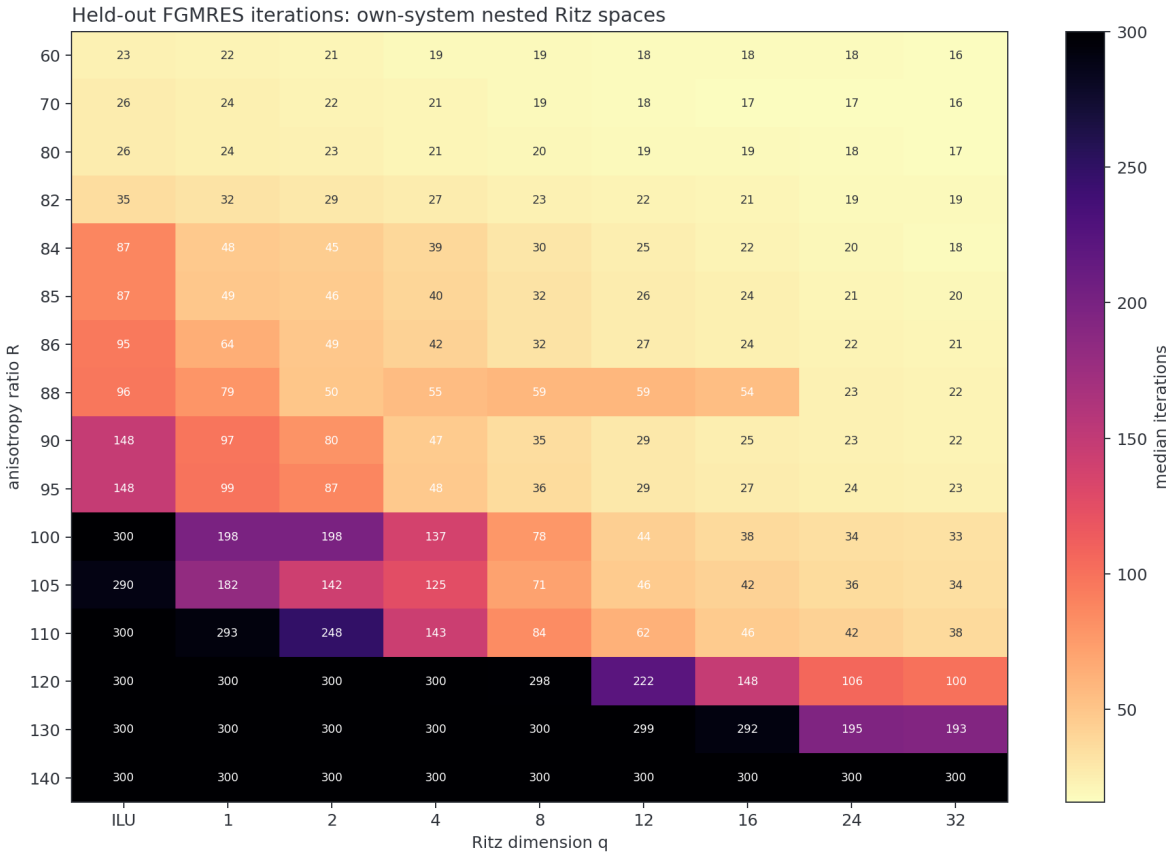


Figure 7: Median iteration count versus anisotropy and Ritz dimension. A fixed small dimension does not cover the full transition.

The system-specific spaces varied systematically in aggregate but were not stable enough for supervised labels. For rank eight, all-pair normalized projector distance had median 0.787, local pairs with $|\Delta R| \leq 2$ had median 0.488, and the median maximum principal angle was 87.4°. Distance in R and projector distance had Spearman correlation 0.884. Nearby reuse retained useful iteration reductions on several pairs, but its hard-regime convergence rate was 0.75 and its median gap to the own oracle was four iterations. Small node perturbations were substantially more damaging than changing the construction RHS.

Tracking individual Ritz directions did not find a globally stable mode. Completing complex conjugate planes reduced neighboring projector distance only from 0.558 to 0.537. Spectral-gap clustering produced no block that was simultaneously stable and solver critical. Low-rank cluster matches had normalized distance 0.845, six clusters were critical, and none also passed the stability gate. Slightly enlarged crossing blocks improved own-system reduction to 68.8% but left neighboring reuse at 75% convergence. Thus the instability was not merely an arbitrary rotation inside a clearly stable rank-eight invariant block.

Transfer also separated right-space information from the test-space partner. Reusing Z_i while rebuilding $A_j Z_i = W_{j \leftarrow i} R_{j \leftarrow i}$ gave, for example, $300 \to 100$ on $95 \to 100$ and $290 \to 94$ on $100 \to 105$, compared with own-oracle counts 78 and 71. Full donor-pair reuse and target-recomputed W behaved similarly. Because the target-side W was already recomputed in the main transfer, poor high-R transfer cannot be blamed on symmetric coupling or reuse of the wrong residual partner.

Finally, 142 controlled spaces and 426 held-out solves tested whether a geometric or short-rollout objective predicts actual acceleration. The solution-space projector metric had Spearman correlation -0.205 with relative iteration reduction. The residual-space metric d_W was stronger overall at -0.840, but aligned in

only two of four perturbation families and had leave-family-out prediction correlation 0.394. Ten-step residual ratio had correlation −0.497 and also reversed ranking on the random-perturbation family. No metric passed the cross-family reliability gate.

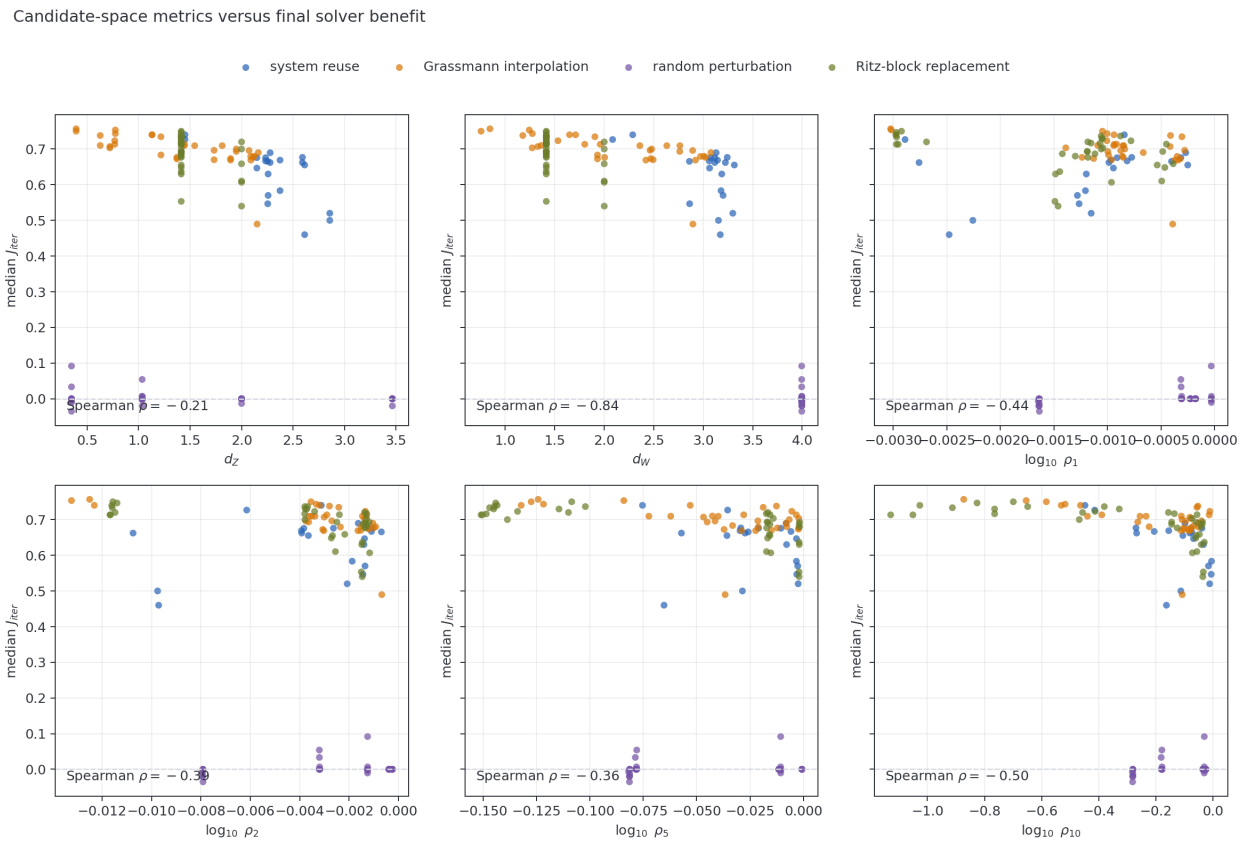


Figure 8: Projector and short-horizon metrics against final iteration reduction. Residual-space geometry is the strongest aggregate diagnostic but not a reliable universal target.

Geometry and solver value can disagree sharply. A donor $R = 90$ space used at $R = 100$ had $d_Z = 2.591$ and $d_W = 3.243$, yet reduced the capped baseline to 97 iterations, giving observed $J_{\text{iter}} = 0.677$. Conversely, controlled perturbations with favorable geometric rank were not uniformly favorable in the solver. Projector similarity is therefore neither necessary nor sufficient for the desired ranking.

6 Direct correction learning

The unstable first Ritz labels motivated a return to direct correction learning, which avoids a fixed low-rank assumption. This branch preserved equation (5). The initial fixed-$N = 900$ MLP took a residual and compact system parameters and predicted the full missing correction. It used one hidden layer of width 64 and 116,228 parameters. On actual solver directions, its median relative error and magnitude ratio were both approximately 996, and all 12 learned solves reached the 800-step cap.

The diagnostic pipeline was then corrected. Actual vectors presented to the ILU preconditioner during Arnoldi were collected on disjoint train, validation, and test trajectories. Homogeneous normalization used $\widehat{v} = v/\|v\|_2$ and trained against $d(v)/\|v\|_2$, with inference $d_\theta(v) = \|v\|_2 N_\theta(\widehat{v}, \xi)$. On held-out ILU trajectories the MLP error fell to 0.770, its norm ratio was 0.923, and cosine similarity was 0.786. After insertion into FGMRES, however, the model changed the Krylov trajectory. The median nearest-training

cosine fell from 0.653 offline to 0.159, closed-loop correction error rose to 2.519, and the norm ratio rose to 2.525. All six MLP solves hit 300 iterations.

A small two-layer, width-16 message-passing GNN used coordinates, normalized vector value, boundary flag, coefficient, stencil edges, relative coordinates, distances, and RBF–FD weights. It had only 3,521 parameters and was permutation equivariant to 2.20×10^{-16}, whereas the MLP equivariance defect was 2.98. Graph structure repaired the ordering defect, but not the solver. The GNN required 134 median iterations and 0.813 s, compared with 11 iterations and 0.0098 s for ILU.

Three rounds of on-policy aggregation stabilized the MLP under its own trajectory: the undamped model moved from no convergence at a 120-step diagnostic cap to 67 iterations with full convergence. Damping αd_θ with $\alpha = 0.1$ reduced the median to 14, and the norm limiter was inactive in the median case. This was a post-hoc mechanism diagnostic, not a deployment result. It remained slower than the 11-iteration, 0.0084-s ILU reference and required 0.020 s online, excluding 18.4 s of training. The GNN remained unreliable. Table 7 and figure 9 summarizes why the direct branch was paused.

Table 7: Direct-complement development. Caps and convergence denominators differ according to the declared experiment and are retained explicitly.

Experiment and method	Correction diagnostic	Iterations	Conv.	Timing comparison
Initial direct MLP	solver-direction error ≈ 996	800 cap	0/12	ILU: 11 iterations
Solver-aware MLP	offline error 0.770; closed-loop 2.519	300 cap	0/6	ILU: 11 iterations
Solver-aware GNN	offline error 1.158	134 median	6/6	0.813 s; ILU 0.0098 s
On-policy MLP, undamped	on-policy error 1.112	67 median	2/2	ILU: 11 iterations
On-policy MLP, $\alpha = 0.1$	damped, no clipping in median	14 median	4/4	0.020 s; ILU 0.0084 s
On-policy GNN	on-policy error 4.464	83.5 median	2/4	0.626 s; ILU 11 iterations

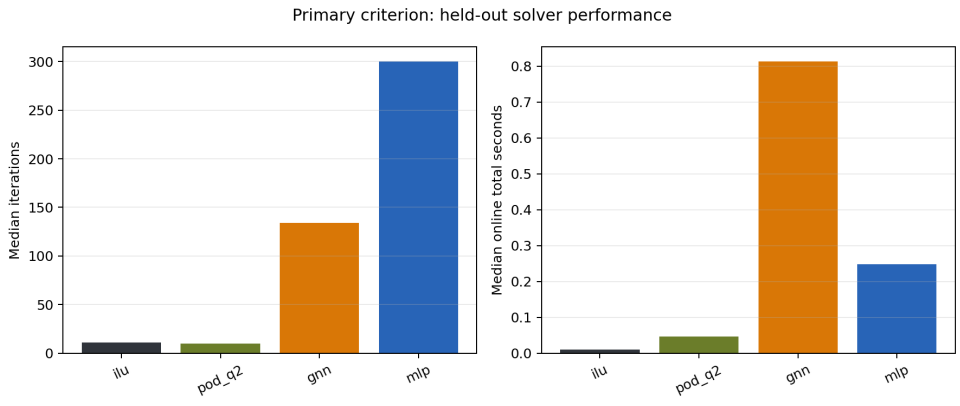


Figure 9: Held-out solver comparison after solver-aware training. Better offline correction metrics did not produce a competitive closed-loop preconditioner.

The target remained $d(v) = A^{-1}v - M^{-1}v$ throughout this branch. Solver-aware sampling fixed the first distribution mismatch, normalization fixed the open-loop scale failure, and on-policy aggregation improved stability. None produced a robust iteration or cost advantage over strong ILU. The evidence gave no reason to increase model size.

7 Recycle-space learning in controlled families

Direct correction learning improved under solver-aware training but remained slower and less reliable than ILU. The next experiments return to recycle spaces within controlled parameter families, where label regularity can be tested before increasing input complexity. The first study used one scalar parameter, fixed nodes, and $R \in [90, 110]$. Training used even ratios, with disjoint odd-ratio validation and test sets. A training-only Z-POD dictionary reached 99.12% energy at $p = 19$. The MLP followed equation (13) and the W-projector loss in equation (14).

The held-out model was solver useful within this restricted family. It converged on every solve and recovered a median 0.939 of the own-oracle iteration reduction. Its normalized d_W remained 0.685, and it needed 89 median iterations. Nearest reuse recovered 0.952 and midpoint Grassmann interpolation 0.983. The learned method matched or improved on nearest reuse for only 20% of test ratios. Deterministic reuse or interpolation therefore remained stronger within this family.

Changing only the dictionary did not repair the ranking. At fixed $p = 19$, projecting held-out oracles into the original Z-POD dictionary produced normalized W error 0.595. The corresponding errors were 0.700 for the training-only AZ-aware dictionary and 0.797 for the joint dictionary. The best learned arm remained the original dictionary with recovery 0.939. This ablation also exposed a sign error in the proposed joint reconstruction expression: minimizing a negative weighted error would reward worse AZ reconstruction, so the implementation explicitly used the intended nonnegative plus-weight objective.

Action supervision changed only the loss. With $AZ = WR$ and $U = ZR^{-1}$, the recycle action on a probe was $c(r) = UW^{\mathsf{T}}r$. Training used 7,743 solver-generated probes from training systems, with 2,999 validation and 3,158 test probes. Median held-out action error improved from 0.935 under W loss to 0.858 under action loss, but the solver worsened from 89 to 98 median iterations and oracle recovery fell from 0.939 to 0.851. Better average action approximation did not preserve the few directions that controlled convergence.

Table 8: Fixed-architecture recycle-space learning and ablations. $\eta = (k_{\mathrm{ILU}} - k_{\mathrm{method}})/(k_{\mathrm{ILU}} - k_{\mathrm{oracle}})$ is computed per RHS before aggregation.

Family/ablation	Reduced rank p	Main offline metric	Median iterations	Median η	Deterministic η
Scalar R, W-projector MLP	19	normalized $d_W = 0.685$	89	0.939	0.983
W-aware dictionary ablation	19	best arm remained Z-POD	89	0.939	0.983
Scalar R, action-loss MLP	19	action error $0.935 \to 0.858$	98	0.851	0.983
Two-parameter (R, θ) MLP	27	normalized $d_W = 0.648$	47	0.921	0.855

The next experiment changed the family rather than the network. It used a spatially constant tensor

$$K(R, \theta) = Q(\theta) \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix} Q(\theta)^{\mathsf{T}}, \qquad Q(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}. \tag{15}$$

Thus the input was finite dimensional, $(R, \theta) \mapsto Z$. A legacy implementation stored the eigenvalues in the opposite order, so the experiment applied the exact conversion $\phi = \theta - 90^\circ$. The maximum tensor mismatch was 3.71×10^{-16}.

The hardness scan covered $R \in \{90, 95, 100, 105, 110\}$ and physical angles 15°–45°, with extra points from 27° to 33°. Angles at most 25° were easy, while angles at least 33° generally capped for both ILU

and rank-eight recycling. Training was therefore restricted, before learning, to the connected transition rectangle $R \in [95, 100]$, $\theta \in [30^\circ, 31^\circ]$ where both classical difficulty and oracle value existed.

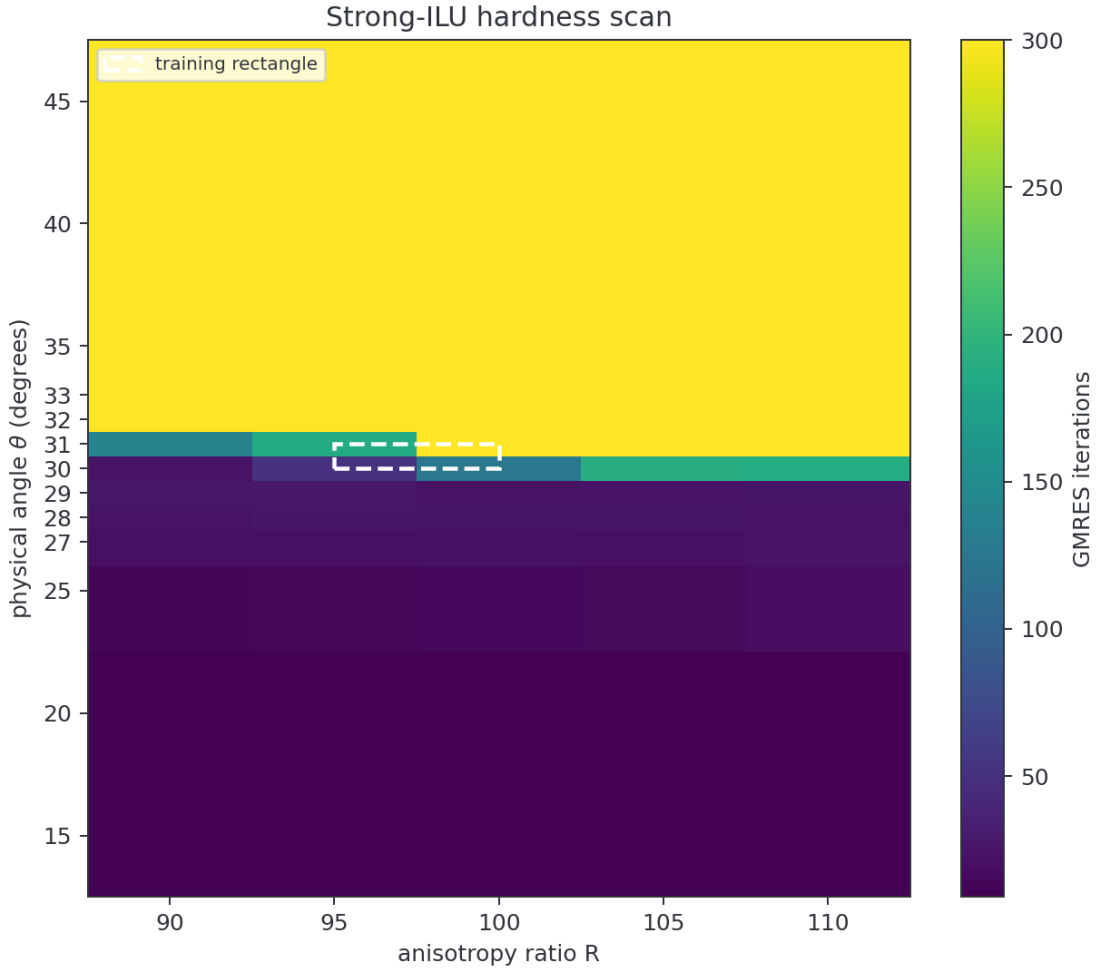


Figure 10: Strong-ILU iteration scan for the spatially constant two-parameter anisotropy tensor.

The training set was a 5×5 tensor grid. Sixteen cell centres were strictly intermediate in both parameters and were split checkerboard-style into eight validation and eight test systems. A training-only dictionary reached 99.1% energy at $p = 27$. The MLP became $2 \to 64 \to 128 \to 216$ and otherwise retained the same optimizer, precision, seeds, and W loss. Deterministic interpolation followed Grassmann geodesics in both coordinate orders and then took their Grassmann midpoint.

On held-out test systems, the selected MLP achieved median recovery 0.921 versus 0.855 for interpolation, with median iteration difference -1.5. The 24 matched RHS solves split 12–12 in wins, the paired Wilcoxon value was $p = 0.074$, and median solve-time difference was -0.001 s. The methods were computationally comparable. The data do not support significant MLP superiority. This is the strongest learned result in the RBF–FD branch, within a narrow fixed-instance interpolation problem.

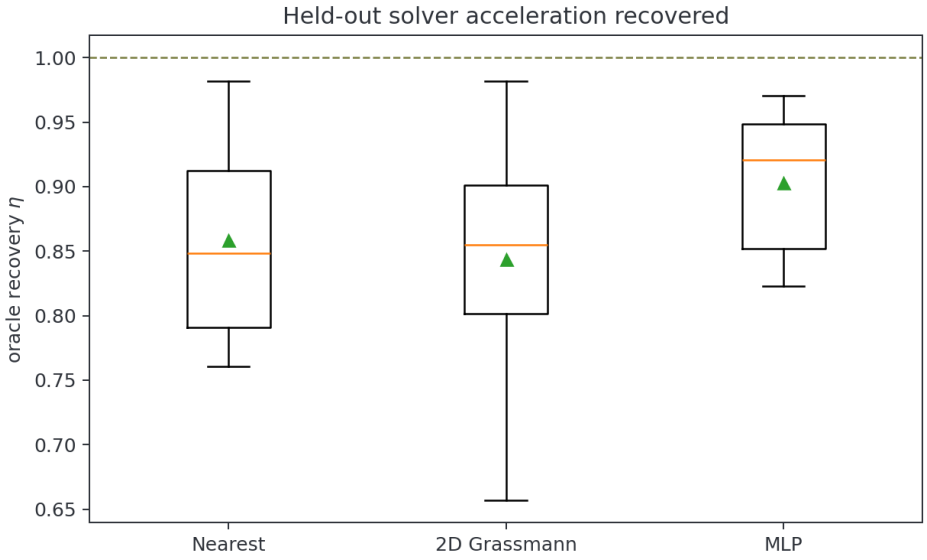


Figure 11: Oracle recovery on the held-out spatially constant (R, θ) family. Heterogeneous pointwise wins preclude a superiority claim.

8 Spatial coefficients break the short fixed-rank oracle

The controlled families showed that learning can interpolate a validated oracle in a narrow setting. The next experiment replaces the finite-dimensional input with a function-valued tensor field and tests whether the same short fixed-rank oracle survives. The tensor is

$$K(x, y) = Q(\theta(x, y)) \operatorname{diag}(R(x, y), 1) Q(\theta(x, y))^{\mathsf{T}}, \tag{16}$$

with

$$\begin{aligned} \log R(x, y) &= a_0 + a_1 \sin(\pi x) + a_2 \cos(\pi y) + a_3 \sin(\pi x) \cos(\pi y), \\ \theta(x, y) &= \theta_0 + b_1 \sin(\pi x) + b_2 \cos(\pi y). \end{aligned} \tag{17}$$

Thirty-six Latin-hypercube fields used seed 20261901. The prescribed ranges were $a_0 \in [\log 80, \log 120]$, $a_1, a_2 \in (-0.18, 0.18)$, $a_3 \in (-0.12, 0.12)$, $\theta_0 \in (25°, 35°)$, and $b_1, b_2 \in (-5°, 5°)$. Realized nodewise ratios spanned 69.10–158.77 and angles 19.46–40.94 degrees. Every field was uniformly SPD and used the exact product-rule operator equation (4).

All 36 discretizations were numerically valid. Manufactured relative error had median 0.0011 and maximum 0.0087. Strong-ILU difficulty was heterogeneous: median iteration count across systems was 27.5 with range 16–300, seven systems needed at most 20, fifteen needed at least 50, and fifteen needed at least 100 or capped. The full distribution was reported before oracle selection.

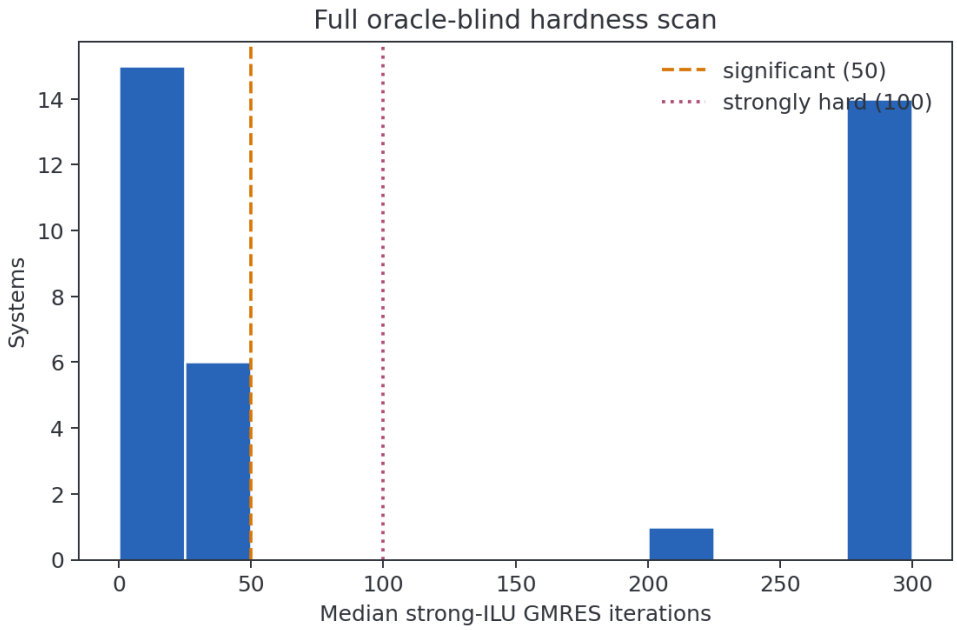


Figure 12: Classical difficulty over the complete 36-field sample.

An 18-system oracle subset was fixed from evenly spaced ILU-difficulty order statistics before any oracle outcome was observed. Among its eight hard systems, only ID 15 passed the strong rank-eight criterion, improving from 211 to 41 iterations. The other seven all remained at the 300-step cap. This failure differs fundamentally from a neural model failing to learn a useful oracle. No model was trained because the oracle itself was not consistently useful.

Table 9: Rank-eight own-system Ritz oracle on the eight hard spatial-field systems.

Field ID	ILU iterations	$q = 8$ Ritz iterations	γ_{oracle}	Oracle convergence
3	300	300	0.000	0.00
7	300	300	0.000	0.00
10	300	300	0.000	0.00
14	300	300	0.000	0.00
15	211	41	0.806	1.00
17	300	300	0.000	0.00
19	300	300	0.000	0.00
34	300	300	0.000	0.00

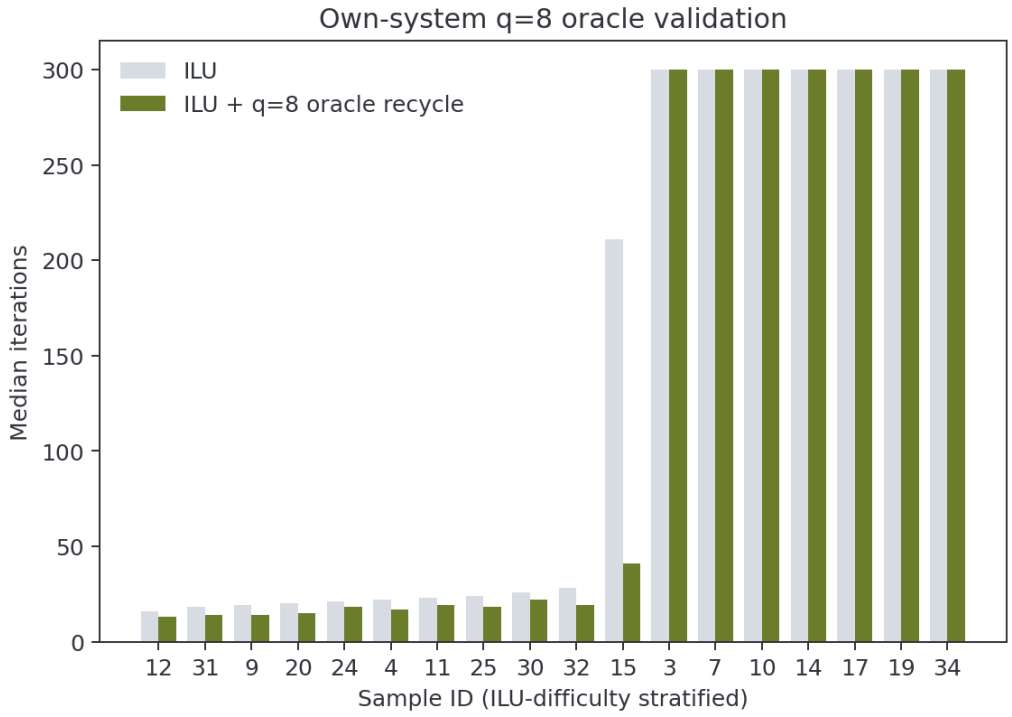


Figure 13: ILU and rank-eight own-oracle iterations across the difficulty-stratified subset.

The oracle spaces were also more complex as a family. Concatenated hard-system Z snapshots required dictionary dimensions 30, 43, and 47 for 95%, 99%, and 99.5% energy. The nearest-field-neighbor normalized projector distance had median 0.825 and the median maximum principal angle was 87.2°. Coefficient-field distance did not explain space distance: Spearman correlation was -0.189 with $p = 0.336$. A controlled diagnostic changed only oracle dimension and definition on the same eight hard systems. Ordinary smallest-magnitude Ritz was tested through requested rank 32 with invariant pairs preserved. ID 15 passed at rank four, and ID 34 joined only at rank 32 with $300 \to 150$. ID 14 improved to 225 without reaching the 40% reduction criterion. IDs 3, 7, 10, and 17 never improved within the first 32 directions. A modest rank increase did not repair the family.

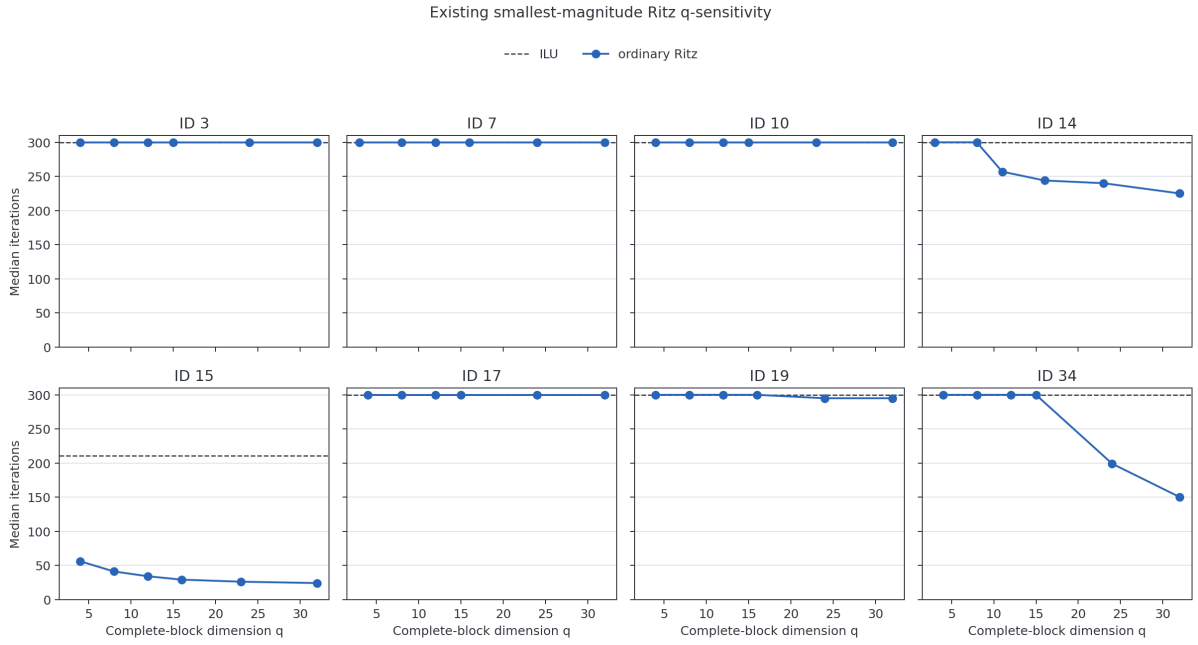


Figure 14: Ordinary Ritz q-sensitivity on the eight hard spatial-field systems.

Harmonic Ritz used the same 64-step right-preconditioned Arnoldi information and selected values nearest zero. Correction POD used three disjoint construction trajectories, 16 actual preconditioner inputs per trajectory, and the unchanged missing action in equation (5). Ordinary and harmonic Ritz each passed two of eight systems. Correction POD passed one.

Table 10: Best tested classical diagnostic spaces on the spatial-field family. Strong success requires all three held-out solves to converge and median reduction at least 40%.

Diagnostic space	Strong successes	Positive control ID 15	Additional success	Interpretation
Ordinary Ritz, $q \leq 32$	2/8	$211 \to 24$ at $q = 32$	ID 34: $300 \to 150$ at $q = 32$	Not a common dimension fix
Harmonic Ritz, $q \leq 32$	2/8	$211 \to 25$ at $q = 32$	ID 34: $300 \to 149$ at $q = 32$	Selection rule does not rescue family
Correction POD, $q \leq 32$	1/8	$211 \to 36$ at $q = 32$	none	Compressibility is not recycle relevance

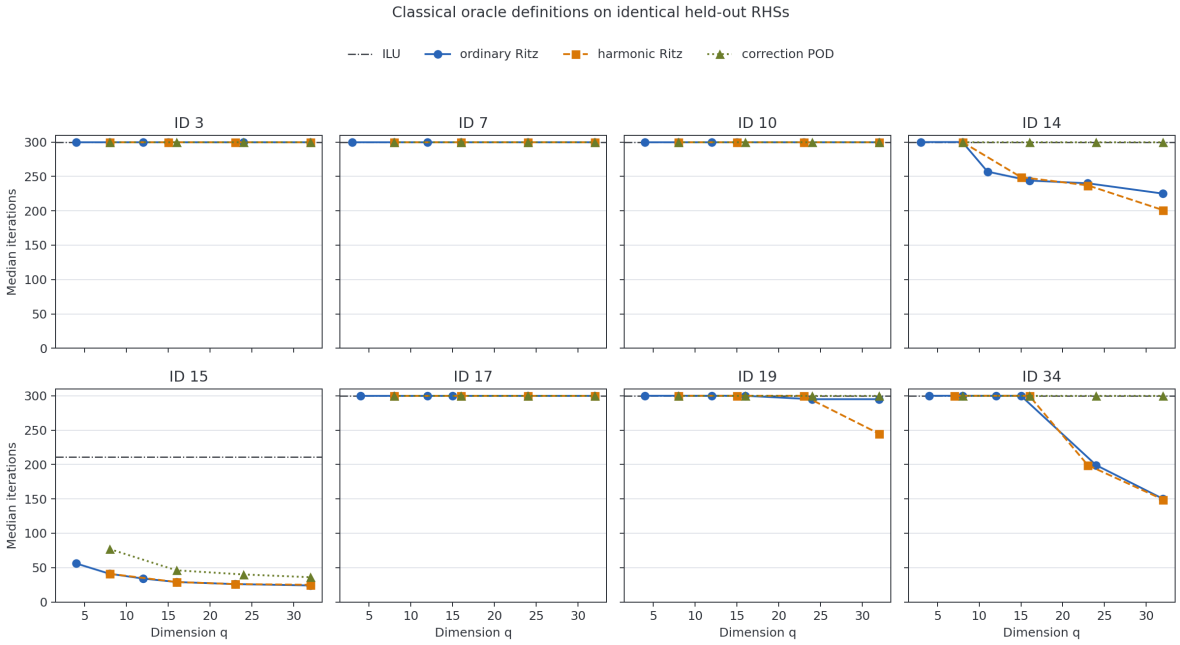


Figure 15: Best ordinary Ritz, harmonic Ritz, and correction-POD outcomes through dimension 32.

The strongest contradiction concerned correction reconstruction and solver relevance. For IDs 17 and 34, correction POD had $q_{95} = q_{99} = 1$. Held-out rank-eight projection errors were 1.76×10^{-4} and 0.005, while both recycled solves reached the 300-step cap. A low-rank missing-action snapshot matrix does not guarantee an effective recycled-GMRES augmentation.

9 Target-specific recycle spaces

Sections 1–8 identified spatially varying anisotropy as the decisive hard regime and showed that the original depth-64, rank-eight construction was inadequate. The next question is direct: if sufficient target-specific Krylov information is available, can it produce a reliable low-dimensional recycle space? This section treats the successful construction as a teacher or oracle. It establishes attainable solver performance before asking whether that information can be transferred or learned.

All systems in this continuation use $N = 5{,}041$, the established RBF–FD discretization, strong ILU with drop tolerance 10^{-3} and fill factor 5, right-preconditioned recycled GMRES, restart 50, and a 300-iteration cap. Construction and evaluation RHSs are disjoint. Convergence requires the true residual

$$\varepsilon_k = \frac{\|b - Ax_k\|_2}{\|b\|_2} \le 10^{-8}. \tag{18}$$

For baseline, candidate, and reference counts k_0, k_Z, and k_*, solver improvement is measured by

$$J_{\text{iter}} = \frac{k_0 - k_Z}{k_0}, \qquad \eta_* = \frac{k_0 - k_Z}{k_0 - k_*}, \qquad k_0 > k_*. \tag{19}$$

A cap remains an observed failure. The reference for η_* is stated in each comparison.

9.1 Depth, rank, and complementary trajectories

For $B = AM^{-1}$ and the Arnoldi relation in equation (7), ordinary Ritz extraction solves $H_m g = \lambda g$. Harmonic Ritz extraction solves

$$\overline{H}_m^{\mathsf{T}} \overline{H}_m g = \lambda H_m^{\mathsf{T}} g. \tag{20}$$

Both constructions preserve real and complex-conjugate blocks and map $V_m g$ through M^{-1}. At depth 64, the maximum relative residual among the first eight ordinary Ritz pairs was 2.65×10^{-5} for the one successful field, ID 15. The other seven original hard fields had values from 0.302 to 0.864. Increasing the Arnoldi depth from 64 to 128 raised the strong-success count at requested rank 32 from two to six of eight. At depth 128, rank 24 already achieved six successes. Harmonic extraction, residual filtering, and spectral clustering did not improve this count. The useful information was spread across several directions.

The fixed depth-128, rank-24 rule passed ten of the full 15-system hard population and four of seven previously unused hard systems. The rank was small relative to N, yet the failure rate prevented a reliable family-level oracle. Multiple construction trajectories were therefore combined. For trajectory ℓ, let $Z^{(\ell)}$ contain at most 24 pair-preserving Ritz directions. Their joint SVD defines

$$\mathcal{Z} = [Z^{(1)}, \ldots, Z^{(s)}] = L \Sigma T^{\mathsf{T}}, \qquad Z_q = L_{:,1:q}. \tag{21}$$

Each prefix is paired through $AZ_q = W_q R_q$ and $U_q = Z_q R_q^{-1}$. Three depth-128 trajectories with requested rank 48 passed all 15 hard systems and all 45 held-out solves. Actual ranks were 32–48 and the median recycled-to-baseline iteration ratio was 0.113.

Table 11: Effects of depth, trajectory count, and retained rank on the 15 hard fields. Success requires convergence on all three RHSs and at least 40% median iteration reduction.

Construction	Arnoldi steps	Requested q	Successes	Median k_Z/k_0
One RHS, $m = 128$	128	24	10/15	0.293
One RHS, $m = 256$	256	24	14/15	0.157
Two RHSs, $m = 128$	256	24	10/15	0.297
Four RHSs, $m = 64$	256	24	1/15	1.000
Three RHSs, $m = 128$	384	48	15/15	0.113

The matched-work comparison in table 11 shows that depth contributes more than same-rank RHS diversity. A single depth-256 trajectory passed 14 systems, whereas two depth-128 trajectories passed ten and four depth-64 trajectories passed one. Retained rank also matters. The depth-256 construction left ID 17 at 228 iterations, while the rank-48 union reduced it to 91. On the original eight fields, truncating the same three-trajectory union from rank 48 to 24 changed IDs 3, 10, and 17 from 33, 38, and 91 iterations to 147, 186, and 300. The data support complementary trajectories only when their added directions are retained.

Requested rank 28 recovered at least 75% of the rank-48 benefit on six of the original eight fields, while rank 32 did so on seven. A first-cycle contraction rule was then validated on 17 hard systems from a fresh 30-field sample. Starting at rank 28 and expanding to the available rank-48 prefix when $\|r_{\text{end}}\|_2 / \|r_{\text{start}}\|_2 \geq 10^{-4}$ produced 51/51 converged solves. Sensitivity was 1.000, specificity was 0.978, and mean final rank fell from 41.8 to 29.8. Its online time exceeded the fixed alternatives, and the full target-specific dictionary still had to be built.

The result of this section is therefore precise. A reliable target-specific recycle space exists. It requires sufficiently deep Arnoldi information and enough retained directions from complementary trajectories. Its own-system construction is expensive, which motivates the transfer and learning experiments that follow.

10 From teacher spaces to neural selection

The own-system teacher resolves the existence question but spends hundreds of target-specific Arnoldi steps. The next task is to reuse information across systems. The investigation moves from direct donor transfer to a global training dictionary, then separates representation from selection, and finally tests the neural map from K to useful dictionary combinations.

10.1 Transfer and the global dictionary

The rank-48 teacher spaces were transferred over all 210 ordered pairs of the 15 hard fields. Each donor Z_i was paired with the target matrix by $A_j Z_i = W_{j\leftarrow i} R_{j\leftarrow i}$, with pairing defects below 4×10^{-13}. Physical nearest-neighbor selection used

$$d_K(i,j) = \frac{\|K_i - K_j\|_{L^2,F}}{\frac{1}{2}(\|K_i\|_{L^2,F} + \|K_j\|_{L^2,F})}. \tag{22}$$

Only two of 15 nearest transfers converged on all RHSs, and median recovery was zero. Outcome-selected best donors converged for nine targets and had median recovery 0.374. Physical distance had correlation 0.023 with recovery. The weak result across the full donor library showed that direct reuse was too restrictive.

A broader representation was built from 43 training teachers in a new 72-field sample. Their concatenation contained 1,456 independent columns. To distinguish representation from selection, factor a candidate dictionary as $A\Psi_p = Q_p R_p$ and let W_* be the target teacher action space. Teacher-informed principal selection forms

$$Q_p^{\mathsf{T}} W_* = L_p \Sigma_p T_p^{\mathsf{T}}, \qquad Z_{p,*} = \operatorname{orth}\big(\Psi_p R_p^{-1} L_{p,:,1:q_*}\big). \tag{23}$$

This is an optimistic representation test because it uses the target teacher. It does not minimize GMRES iterations.

Table 12: Teacher-informed selection on seven hard test systems. Final ranks match the target teachers. Recovery is relative to the full own-system teacher.

Candidate	Converged RHSs	Median iterations	Median η_*	Median d_W
Solution-energy $p = 168$	12/21	197	0.376	0.692
Residual-aware $p = 168$	18/21	96	0.747	0.537
Residual-aware $p = 428$	18/21	66	0.854	0.403
Residual-aware $p = 573$	18/21	77	0.820	0.356
Full training span $p = 1456$	21/21	36	0.974	0.149
Own-system teacher	21/21	31	1.000	0.000

The rank-168 solution-energy basis captured 99.02% of training energy yet converged on only 12/21 hard-test RHSs. An action-aware basis used paired blocks $A_i U_i = W_i$. With $\mathcal{U} = [U_i]$, $\mathcal{W} = [W_i]$, and the leading right singular vectors T_p of $\mathcal{W}$, it formed

$$\Psi_p^{\mathrm{act}} = \operatorname{orth}(\mathcal{U} T_p), \qquad T_p \in \operatorname*{argmin}_{T^{\mathsf{T}} T = I_p} \left\| \mathcal{W} - \mathcal{W} T T^{\mathsf{T}} \right\|_F^2. \tag{24}$$

This improved the rank-168 ceiling to 18/21. Larger action-aware bases improved average geometry, but system 63 remained capped through $p = 573$. The full 1,456-column training span reached 21/21 convergence and median 36 iterations. The training information is therefore sufficient under teacher-informed selection. Energy truncation loses low-energy directions that matter to the solver.

10.2 The neural mapping remains unreliable

The central neural problem is now isolated:

$$K \longmapsto C_\theta(K) \longmapsto Z_\theta = \operatorname{orth}\big(\Psi C_\theta(K)\big). \tag{25}$$

An initial rank-143 dictionary and both a parameter MLP and field-input FNO produced median 300-iteration solves on the seven hard test systems. The full teacher converged on all 21 RHSs. Expanding to

the full dictionary removed the representation limit but did not solve selection. A coefficient-loss MLP had training $d_W = 0.961$. Residual-projector loss improved training d_W to 0.698 for the MLP and 0.769 for the FNO, while both methods still converged on only three hard-test RHSs. The fitting failure already appeared on training systems.

Donor-mixture prediction reduced the output dimension, while compression of selected unions again removed useful directions. The selected unions supported a teacher-informed result of 21/21 convergence, median 78 iterations, and recovery 0.792. Target-action energy compression fell to 3/21. A depth-64 Krylov-overlap reducer improved this to 18/21 and median 144 iterations. Short solver-response selection used substantially more probe work and performed worse. These controls reinforce the main result and do not define separate methods.

The full dictionary Ψ contains sufficient target information under teacher-informed selection. The learned mapping from K to useful combinations of Ψ remains unreliable. This failure motivates a change of input. Short target-local Krylov information may provide the selector with the operator response that the coefficient field alone did not supply.

11 Classical selection from local probes

Neural selection from K failed even when the training dictionary could represent effective spaces. The project therefore moved temporarily to a classical diagnostic: use a short target-local Krylov probe and select dictionary combinations by overlap and SVD. This experiment asks whether local operator response makes selection easier before another neural formulation is attempted.

Let $A\Psi = Q_\Psi R_\Psi$ and let S contain target-local probe information. Replacing the unavailable W_* in equation (23) gives

$$Q_\Psi^{\mathsf{T}} S = L\Sigma T^{\mathsf{T}}, \qquad Z_q^{\mathrm{probe}} = \mathrm{orth}\big(\Psi R_\Psi^{-1} L_{:,1:q}\big). \tag{26}$$

One depth-64 or depth-128 Arnoldi trajectory produced 18/21 convergence and median 91 or 96 iterations under the inherited rank policy. Three depth-64 trajectories gave the same convergence. Retaining 24 harmonic Ritz directions per trajectory defines

$$S_{\mathrm{h}} = \mathrm{orth}\big([AZ_{\mathrm{h}}^{(1)}, AZ_{\mathrm{h}}^{(2)}, AZ_{\mathrm{h}}^{(3)}]\big), \qquad \dim S_{\mathrm{h}} = 72. \tag{27}$$

The main result is that this classical construction succeeds on six of the seven diagnostic hard systems. All three failed evaluation RHSs belong to system 63. The other systems mainly needed a larger retained prefix. Keeping all 48 ordered directions reduced the median from 89 iterations under the inherited adaptive policy to 33 without changing the probes.

Table 13: Nested prefixes of the harmonic-probe construction. All remaining failures at rank 24 or above belong to system 63.

Retained q	Converged RHSs	Median iterations	Median η_*
8	3/21	300	0.000
16	9/21	300	0.000
24	18/21	104	0.731
32	18/21	48	0.923
40	18/21	40	0.960
48	18/21	33	0.985

The rank response was broad for five targets. System 43 had one important block at directions 25–32, which saved 75 iterations when added. A separate eight-step rank controller stopped that system at rank

eight and reduced total convergence to 15/21. Its diagnostic work also raised the seven-target total from 2,825 to 4,119 iteration equivalents. The earlier adaptive rule had been validated on a different ordered dictionary and did not transfer to this one.

To describe the remaining failure, define target containment for orthonormal Q and W_* by

$$\mathcal{C}(Q, W_*) = \frac{\left\|Q^\mathsf{T} W_*\right\|_F^2}{q_*}. \tag{28}$$

The eigenvalues of $W_*^\mathsf{T} Q Q^\mathsf{T} W_*$ are squared principal cosines, with zeros for missing dimensions. System 63 had ten teacher combinations with coverage below 0.5 in the harmonic probe space. Removing any one preserved teacher convergence, while a six-direction augmentation only restored convergence at 229 iterations. The failure was collective rather than a single missing mode.

The teacher complement to the fixed rank-48 probe pair (U_c, W_c) is obtained from

$$E_c = (I - W_c W_c^\mathsf{T}) W_\mathrm{o} = L_c \Sigma_c T_c^\mathsf{T}, \qquad W_{\Delta,r} = L_{c,:,1:r}, \tag{29}$$

with the solution partner formed from the corresponding compensated teacher block. Rank-eight and rank-twelve oracle complements restored all 21 solves. Rank twelve reduced system 63 from the cap to 95 iterations. This result identifies a compact missing subspace and leads directly to the next question: did the short probe fail to contain that information, or did harmonic compression discard it?

12 Raw Krylov information and the system-63 exception

Section 11 resolved six hard systems and left one repeated diagnostic exception. This section follows that system-63 branch. Its conclusions identify a possible robustness mechanism, though the frequency of that mechanism on fresh hard systems remains unknown.

Combined depth-64 raw Arnoldi spaces had median containment 0.740 for the unresolved correction, compared with 0.438 after harmonic compression. On system 63 the values were 0.901 and 0.258. The raw probe therefore contained much of the missing information, which was lost during spectral compression.

To preserve the solution and action pairing, stack the three Arnoldi solution blocks and their actions as Y and $F = AY$. Remove the fixed classical pair by

$$Y_\perp = Y - U_c W_c^\mathsf{T} F, \qquad F_\perp = (I - W_c W_c^\mathsf{T}) F = A Y_\perp, \tag{30}$$

then compute the rank-revealing factorization

$$F_\perp \simeq Q_\mathrm{raw} \Sigma_\mathrm{raw} T_\mathrm{raw}^\mathsf{T}, \qquad X_\mathrm{raw} = Y_\perp T_\mathrm{raw} \Sigma_\mathrm{raw}^{-1}, \qquad A X_\mathrm{raw} = Q_\mathrm{raw}. \tag{31}$$

Singular values below 10^{-10} of the largest are discarded. For correction rank r, the teacher-informed raw-space selector chooses

$$C_{*,r} \in \operatorname*{argmax}_{C^\mathsf{T} C = I_r} \left\| C^\mathsf{T} Q_\mathrm{raw}^\mathsf{T} W_{\Delta,r} \right\|_F^2. \tag{32}$$

The raw-space oracle in table 14 converged on all 21 RHSs and reduced system 63 to 92 iterations. The needed information was present in the short raw probe. Neural mixing failed to extract it. At rank twelve, probe-only training containment was 0.201 against a raw-oracle value of 0.446. Adding K reduced it to 0.113. The large training gap reveals a representation or optimization failure before generalization.

The first neural scorer also depended on the arbitrary basis of the raw candidate. Under an orthogonal coordinate change O,

$$(X_\mathrm{raw}, Q_\mathrm{raw}) \mapsto (X_\mathrm{raw} O, Q_\mathrm{raw} O), \qquad P_r \mapsto O^\mathsf{T} P_r O. \tag{33}$$

Table 14: Rank-12 corrections with the same fixed classical prefix. The last two columns give system-63 results and convergence over its three RHSs.

Method	Converged RHSs	Median iterations	ID 63 iterations	ID 63 η_*
Fixed classical $q = 48$	18/21	33	300 [0/3]	0.000
Exact complementary oracle	21/21	26	95 [3/3]	0.785
Harmonic local correction	21/21	27	298 [3/3]	0.008
Raw-space oracle	21/21	25	92 [3/3]	0.800
Raw probe-only neural mixing	21/21	31	235 [3/3]	0.251
K plus raw-probe neural mixing	20/21	32	299 [2/3]	0.004

The physical correction should remain fixed. The scorer had median physical prediction distance 0.926 under this test. A basis-equivariant replacement predicted a symmetric coordinate operator and used its leading eigenspace. Its 33-word span contained six primitive operators, symmetric quadratic products, and primitive cubes. The primitives represented singular-value energy, trajectory participation, and low- and high-frequency spatial Gram matrices. This construction reduced the equivariance defect to about 10^{-8}, yet rank-12 training containment fell to 0.072.

Per-system coefficient fitting improved hard-test containment to 0.274, far below the raw-oracle value 0.788. Direct Frobenius projection of the target projector onto the same operator span gave median residual energy 0.807 at rank twelve. A six-word cross-RHS spatial extension changed this to 0.806. The tested equivariant representation therefore lacks the required projector structure. Reweighting the same features or adding similar low-order words has little support.

This branch was driven heavily by system 63. It shows that harmonic compression can discard a compact solver-relevant complement and that coordinate symmetry alone cannot recover it. The frequency of this mechanism in a fresh hard population remains unknown.

13 Current Issues and Next Directions

The evidence now forms a clear sequence. Target-specific recycle spaces exist, but their construction is expensive. The full training dictionary contains effective target information, while neural selection from K is unreliable. Short target-local Krylov information makes classical selection much easier and solves six of seven diagnostic hard systems. The remaining system-63 case reveals a deeper missing-complement mechanism, though its prevalence is unknown.

The primary next direction should therefore reconsider the neural formulation. Instead of mapping K directly to a recycle space, use

$$(K, \text{ very short target-local Krylov probe}) \longmapsto \text{neural selector} \longmapsto \text{useful combinations of } \Psi. \tag{34}$$

The network should reduce the Krylov information needed for reliable selection. Its value must be tested at equal probe depth against two controls: classical short-probe selection and a probe-only neural selector. A useful model would allow a depth such as $m = 8$ or $m = 16$ to approach the performance of a substantially deeper classical probe. The current results do not yet support that outcome. They define a focused comparison that can establish whether K adds predictive value once local operator response is available.

Efficiency remains part of the scientific question. Each comparison should report iteration count together with probe applications, dictionary and pairing cost, prediction cost, and total online time. For n_{rhs} repeated solves, the relevant accounting is

$$T(n_{\mathrm{rhs}}) = T_{\mathrm{ILU}} + T_{\mathrm{probe}} + T_{\mathrm{pair}} + T_{\mathrm{prediction}} + n_{\mathrm{rhs}} T_{\mathrm{solve}}. \tag{35}$$

Teacher generation and training are offline costs and should be reported separately. Success requires a favorable shift in the probe-cost and solver-performance tradeoff after prediction cost is included.

Further raw-space and equivariant work should remain secondary. First sample fresh hard fields and measure how often a fixed rank-48 harmonic-probe space misses a complement comparable to system 63. A dedicated robustness mechanism is justified only if that failure recurs often enough to affect overall solver reliability. The present 33-word operator family should be retired, since both per-system fitting and direct projector projection show a large representation deficit. A richer equivariant architecture becomes relevant only after the prevalence test and only if it can be evaluated against the same short-probe controls.

Any surviving method should be frozen before final evaluation on new coefficient fields with oracle-blind hardness selection and disjoint construction and evaluation RHSs. The decisive result is robust total solution cost relative to strong ILU and the best classical probe selector. Teacher-projector similarity remains a diagnostic rather than the final objective.

References

[1] David Amsallem and Charbel Farhat. Interpolation method for adapting reduced-order models and application to aeroelasticity. *AIAA Journal*, 46(7):1803–1813, 2008.

[2] Eric de Sturler. Nested Krylov methods based on GCR. *Journal of Computational and Applied Mathematics*, 67(1):15–41, 1996.

[3] Alan Edelman, Tomás A. Arias, and Steven T. Smith. The geometry of algorithms with orthogonality constraints. *SIAM Journal on Matrix Analysis and Applications*, 20(2):303–353, 1998.

[4] Michael Koch, Sabine Le Borne, and Willi Leinen. Smaller stencil preconditioners for linear systems in RBF-FD discretizations. *Numerical Algorithms*, 98:1313–1335, 2025.

[5] Jian Luo, Jie Wang, Hong Wang, Huanshuo Dong, Zijie Geng, Hanzhu Chen, and Yufei Kuang. Neural Krylov iteration for accelerating linear system solving. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

[6] Michael L. Parks, Eric de Sturler, Greg Mackey, Duane D. Johnson, and Somnath Maiti. Recycling Krylov subspaces for sequences of linear systems. *SIAM Journal on Scientific Computing*, 28(5):1651–1674, 2006.

[7] Youcef Saad. ILUT: A dual threshold incomplete LU factorization. *Numerical Linear Algebra with Applications*, 1(4):387–402, 1994.